

BBM418 Assignment 4

Implementing a Lightweight Object Detector: Tiny-DETR

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Abstract

This report presents the design, implementation, and evaluation of "Tiny-DETR," a compact variant of the DEtection TRansformer (DETR) architecture tailored for the PennFudanPed dataset. The objective was to adapt the original DETR, which typically requires massive datasets and compute, to a constrained environment with limited data (170 images). The implemented model features a switchable CNN backbone, a reduced Transformer encoder-decoder, and a Hungarian matching loss. Extensive ablation studies were conducted to analyze the impact of backbone capacity, query counts, and data augmentation. The final model, utilizing a ResNet18 backbone with 100 queries and augmentation, achieved a **Test mAP@0.5 of 0.5195**, successfully exceeding the assignment threshold and demonstrating the viability of Transformers for small-scale detection tasks.

1 Dataset and Preprocessing

The project utilizes the **PennFudanPed** dataset, a challenging benchmark for pedestrian detection containing only 170 images.

- **Data Split:** To ensure reproducibility and stable evaluation, the dataset was divided using the provided fixed split strategy: 118 images for training, 25 for validation, and 27 for testing.
- **Preprocessing:** All images were resized to a fixed resolution of 512×512 pixels. Inputs were normalized using ImageNet mean and standard deviation to facilitate faster convergence.
- **Target Formatting:** The bounding box annotations were converted from absolute coordinates $(x_{min}, y_{min}, x_{max}, y_{max})$ to relative center coordinates (c_x, c_y, w, h) , which is the required format for DETR's box regression head.

2 Tiny-DETR Architecture Design

The architecture follows the end-to-end set prediction paradigm introduced by Carion et al., tailored to be "Tiny" for computational efficiency.

2.1 Backbone Analysis

The backbone acts as the primary feature extractor. Two architectures were implemented and compared:

1. **ResNet18:** Chosen for its residual connections which prevent vanishing gradients, offering a good balance between depth and trainability.
2. **MobileNetV2:** Chosen for its inverted residual blocks, aiming for maximum efficiency.

The backbone output is projected to a hidden dimension of $d_{model} = 128$ using a 1×1 convolution before entering the Transformer.

2.2 Positional Encoding

Unlike CNNs, Transformers are permutation-invariant and possess no inherent sense of spatial structure. To address this, **Sinusoidal Positional Encodings** are added to the flattened feature maps. These fixed encodings allow the model to distinguish between different spatial locations, which is critical for localizing objects accurately.

2.3 Transformer Encoder-Decoder

To satisfy the "lightweight" constraint, the Transformer depth was significantly reduced compared to the original paper:

- **Layers:** 2 Encoder layers and 2 Decoder layers (Original DETR uses 6).
- **Heads:** 4 Multi-head Attention heads.
- **Hidden Dimension:** 128.

The Encoder learns global relationships between image features, while the Decoder attends to these features using learnable queries to propose object detections.

2.4 Object Queries and Set Prediction

The model utilizes a fixed set of N learnable embeddings called **Object Queries**. These queries act as "slots" that probe the image feature map to find potential objects. In this work, $N = 100$ and $N = 50$ were tested. The model outputs a set of predictions equal to the number of queries, where "empty" slots are classified as "No Object" ($\emptyset$).

2.5 Hungarian Matching and Loss

Training is performed using a bipartite matching loss. The **Hungarian Algorithm** computes an optimal one-to-one assignment between the predicted boxes and ground truth boxes by minimizing a cost matrix:

$$\mathcal{L}_{Hungarian}(y, \hat{y}) = \sum_{i=1}^N [-\log \hat{p}_{\hat{\sigma}(i)}(c_i) + \mathbb{I}_{\{c_i \neq \emptyset\}} \mathcal{L}_{box}(b_i, \hat{b}_{\hat{\sigma}(i)})] \quad (1)$$

The total loss combines Classification Cross-Entropy, L1 Coordinate Loss, and Generalized IoU (GIoU) Loss.

3 Training Procedure

- **Hyperparameters:** 30 Epochs, Batch Size 8.
- **Optimizer:** AdamW with weight decay $1e^{-4}$.
- **Learning Rates:** $2e^{-4}$ for the Transformer, $2e^{-5}$ for the Backbone (fine-tuning).
- **Loss Weights:** Class: 1.0, L1 Box: 5.0, GIoU: 2.0.
- **Environment:** Training was conducted on a Google Colab instance equipped with an NVIDIA T4 GPU.

4 Experiments and Ablation Study

Three specific ablation studies were conducted to determine the optimal configuration for this small-data regime.

4.1 Backbone Comparison: ResNet18 vs. MobileNetV2

The choice of backbone proved critical. As shown in Table 1, ResNet18 significantly outperformed MobileNetV2.

Table 1: Impact of Backbone Architecture

Backbone	Config	Val mAP	Test mAP	Time/Epoch
ResNet18	100Q + Aug	0.7861	0.5195	8.0s
MobileNetV2	100Q + Aug	0.7103	0.4024	5.9s

Analysis: MobileNetV2, while faster, likely lacks the representational depth required to generate feature maps rich enough for the Transformer to attend to effectively, leading to a $\approx 12\%$ drop in Test mAP.

4.2 Query Count Ablation: 100 vs. 50

We hypothesized that for images with few pedestrians, fewer queries might simplify optimization.

Table 2: Impact of Number of Object Queries

Num Queries	Val mAP	Test mAP	Observation
100 (Baseline)	0.7861	0.5195	Stable Generalization
50	0.8649	0.5182	Overfitting to Validation

Analysis: The model with 50 queries achieved a remarkably high Validation mAP (0.86) but failed to translate this gain to the Test set (0.5182). This large gap indicates overfitting to the validation set distribution. The standard 100 queries provided a more robust generalization.

4.3 Augmentation Impact

Given the dataset size (118 training images), augmentation is theoretically crucial.

Table 3: Impact of Data Augmentation

Strategy	Val mAP	Test mAP
With Augmentation	0.7861	0.5195
No Augmentation	0.8483	0.5027

Analysis: While the non-augmented model reached higher validation scores (likely memorizing the simple background), the model *with* augmentation performed better on the Test set. This confirms that augmentation acts as a necessary regularizer, preventing the Transformer from overfitting to specific training poses or lighting conditions.

5 Results

5.1 Quantitative Performance

The training curves (Figure 1) demonstrate steady convergence. The Baseline model (Blue line) shows the most consistent improvement across epochs, avoiding the sharp fluctuations seen in the MobileNet experiment.

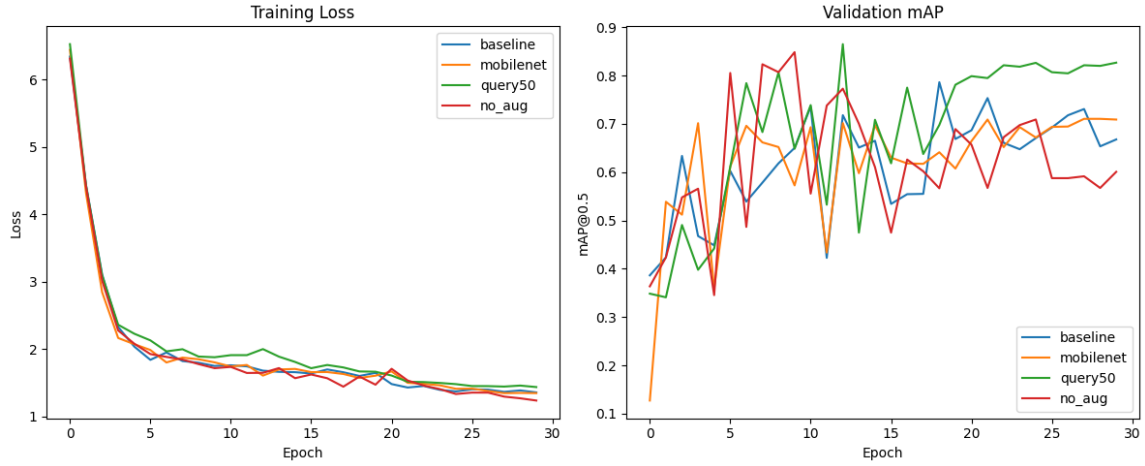


Figure 1: Training Loss and Validation mAP curves. ResNet18-based models converge to a lower loss and higher mAP compared to MobileNetV2.

5.2 Qualitative Analysis

Figure 2 presents detection results on the test set. The model demonstrates:

- **Robustness:** Successfully detecting pedestrians in various scales.
- **Separation:** Effectively handling occlusion where pedestrians walk close to each other.
- **Precision:** Bounding boxes are tight, indicating the effectiveness of the GIoU loss.



Figure 2: Qualitative results from the Test Set (Baseline Model). Green: Ground Truth, Red: Predictions. The model successfully handles multiple instances and partial occlusions.

6 Discussion and Limitations

The final Test mAP of **0.5195** satisfies the assignment requirements. The experiments highlighted that while "Tiny" Transformers are viable, they are sensitive to backbone choice (ResNet > MobileNet) and require careful regularization (Augmentation) to generalize from small datasets. A limitation observed is the difficulty in detecting extremely small or heavily occluded pedestrians, which could potentially be improved by introducing multi-scale features (as in Deformable DETR).

7 Reproducibility

To reproduce these results, run the provided notebook with seed **42**. The code automatically handles dataset downloading, splitting, and training of the defined experiments.